



A Project Synopsis

on

BundleWise: A Retail Analytics Tool for Smart Bundling and Profit-Driven Pricing

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For the course

Course Code: MDS481

Course Name: PROJECT-I

1. Introduction

a. About the Domain

The field of retail analytics focuses on analyzing data produced by retail operations in order to assist with operational and strategic decision-making. To assist merchants in making wise decisions, it entails analyzing trends in sales, consumer behavior, inventory movement, and overall company performance.

The retail industry is changing quickly in the current digital era as a result of growing competition, shifting consumer preferences, and the expansion of online purchasing. As a result, retailers are depending more on data to better understand customer wants, boost profitability, and improve service quality.

Retail analytics helps businesses identify what products are selling, who their customers are, when and how purchases are made, and how to respond to changing market dynamics. It supports better planning, targeted marketing, efficient stock management, and improved customer engagement.

Retail analytics is essential for helping traditional and online retailers stay competitive, flexible, and customer-focused in a fast-paced industry by turning unstructured business data into insightful knowledge.

b. Motivation

Businesses are always searching for methods to boost productivity, boost profitability, and gain a deeper understanding of consumer behavior in the competitive retail industry. Effective use of their current sales data is difficult for many merchants, particularly small and mid-sized businesses. In order to estimate demand, evaluate client buying patterns, or determine strategic pricing, they frequently lack the technical understanding or advanced tools necessary.

Another key challenge is that product purchase data is not always recorded in complete transaction bundles. It may be more difficult to determine related products or client intent when items are bought throughout multiple visits. These useful time-based trends are not shown by conventional techniques that depend on single-invoice analysis.

Additionally, sales trends and consumer preferences can be greatly impacted by outside variables like festivals, seasonal shifts, or promotional events. Retailers run the risk of overstocking or missing out on sales opportunities if they are unable to predict and adjust to these variations.

This project is driven by the need to build an intelligent and user-friendly solution that can help retailers:

- Discover meaningful product relationships across multiple transactions,
- Anticipate demand by analyzing past trends and external factors,
- Make data-informed pricing and bundling decisions,
- Enhance customer satisfaction and business outcomes.

The system wants to make advanced retail insights available to all kinds of merchants, regardless of their technical expertise, by combining automation, predictive analytics, and user-friendly design.

c. Commonly Used Techniques in Retail Analytics

Technique	Purpose
Descriptive Analytics	Summarizes past data, like sales reports or customer behavior.
Predictive Analytics	Forecasts future trends such as demand or sales volume.
Prescriptive Analytics	Recommends actions like pricing or inventory decisions.
Market Basket Analysis	Identifies products frequently bought together.
Sequential Pattern Mining	Discovers time-based purchase sequences across visits.
Clustering/Segmentation	Group customers or products based on similar characteristics.
Dynamic Pricing	Adjusts product prices based on demand and seasonal trends.

2. Objectives

a. Primary Objectives

The main objective of this project is to develop a **web-based application** that enables retailers to discover smart product bundles and optimize their pricing based on purchasing patterns, forecasted sales trends, and seasonal demand

b. Secondary Objectives

- Enable retailers to upload sales data in common formats (e.g., CSV) for seamless processing.
- Incorporate seasonal trends, holidays, and local events into forecasting and pricing strategies.
- Ensure the system is easy to use, requiring no technical background to operate.
- Make the platform scalable and efficient to support different retail business sizes.
- Offer intuitive visualizations to support quick business decision-making.
- Provide options to export analysis results and pricing suggestions for business reporting.

3. About the Project

This project presents a smart retail analytics platform that helps retailers make data-driven decisions by analyzing their sales and product data. The system is designed to work even when the data does not contain complete invoice-level item groupings, making it suitable for small and medium-sized retailers with scattered or incomplete transaction records.

Retailers can upload their historical sales data to the system in a simple format (e.g., CSV with customer ID, product, and date). The system then performs **Sequential Pattern Mining** to discover frequently purchased product sequences over time. These sequences reveal product combinations that are commonly bought across multiple visits or transactions, enabling effective bundling strategies.

Once item sequences are identified, the system uses **time series forecasting** models to predict future sales and profit trends. Forecasts consider seasonal effects and external factors like holidays or local events, improving their accuracy and relevance.

In the final stage, the system applies **dynamic pricing logic** to recommend price adjustments for individual products or product bundles. These suggestions may include offering discounts, creating upsell packages, or down-selling low-performing items to clear inventory.

All findings and recommendations are presented through an **interactive dashboard**, making it easy for retailers to explore sequences, visualize trends, and apply pricing decisions. The dashboard supports file upload, forecast visualization, bundle suggestions, and report generation. The system is built to be scalable, intuitive, and usable even by retailers with limited technical knowledge.

a. Tools and Technologies Used

- **Python** – Core programming language used for development.
- **Pandas and NumPy** – For data preprocessing and manipulation.
- **PrefixSpan / SPADE (via SPMF or mlxtend)** – For Sequential Pattern Mining (SPM).
- **ARIMA, Facebook Prophet, LSTM (Keras/TensorFlow)** – For sales and profit forecasting.
- **SciPy** – For pricing optimization using linear programming and numerical analysis.

- **Streamlit / Dash / Plotly** – To build an interactive dashboard for visualizing insights.
- **Flask / Streamlit Cloud / Render** – For optional deployment and access over the web.
- **CSV file handling** – Used for inputting transaction data.

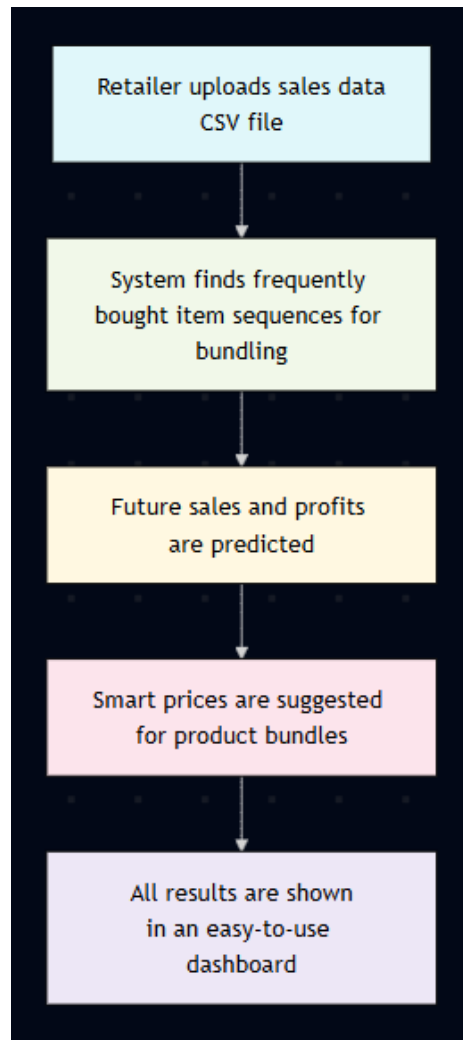
b. Input

- Historical sales data
- Product data

c. Output

- List of products that are often bought together over time
- Predictions of future sales and profits (weekly or monthly)
- Suggested product bundles based on customer buying patterns
- Price recommendations for products and bundles
- Easy-to-use dashboard showing all results and insights

d. Block Diagram



4. Expected Outcome

- A web-based system will be developed to identify frequently purchased product sequences, forecast sales and profits, recommend dynamic pricing for product bundles, and present all insights through an interactive, user-friendly dashboard, helping retailers understand customer buying behavior and make data-driven decisions.
- It can be developed into an academic research article for conferences or journals.
- This project offers **potential for patent filing** due to its unique analytical approach.
- It is scalable as a **commercial web app** for real-world retail use.
- All original work can be applied for **possible copyright protection** to ensure legal ownership.

5. Timeline

Phase	Duration	Activities
Requirement Gathering & Planning	Week 1	Define project scope, finalize dataset structure, identify tools and tech stack.
Data Collection & Preprocessing	Week 2 - Week 3	Collect sample data, clean and structure it, and preprocess it.
Sequential Pattern Mining	Week 4 - Week 5	Implement an algorithm to find frequent product sequences.
Forecasting Module Development	Week 6 - Week 7	Build models to forecast sales and profits.
Pricing Optimization	Week 8	Recommend dynamic prices for products and bundles.
Dashboard & UI Development	Week 9 - Week 10	Build an interactive dashboard to display all insights.
Testing & Validation	Week 11	Test all modules and validate results.
Final Report & Deployment	Week 12	Test all modules and validate results.

6. References

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